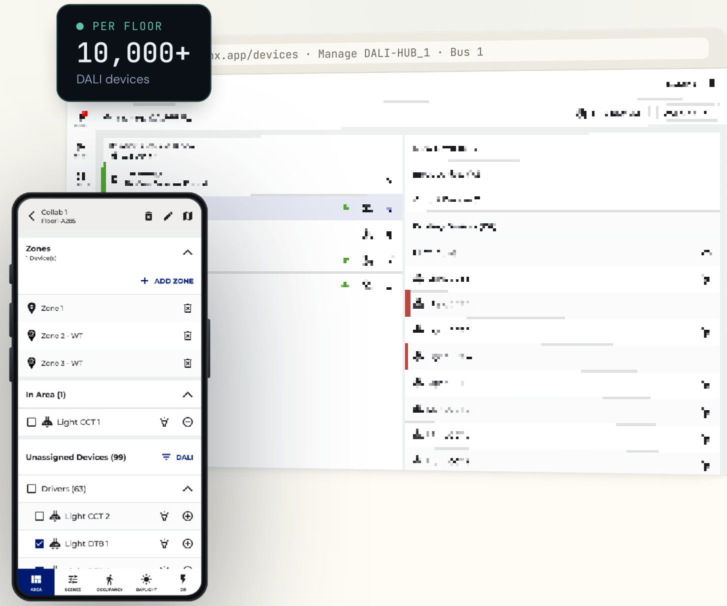
COOPER LIGHTING · SIGNIFY —— PETER GILGAN MISSISSAUGA HOSPITAL

Designing a product that didn't exist, for a protocol that *couldn't change.*



DALI-2 Lighting System, an industry-first commissioning experience for a protocol-constrained commercial IoT product, taken from zero to shipped across hardware, firmware, and field reality at once. First deployment: the largest DALI-2 install in North America.

Enterprise IoT · Healthcare · Industry First · First Release Shipped

ROLE	SCOPE	TEAM
Lead Product Designer	Research → Delivery	15 cross-functional

PLATFORM

Mobile · Web

KEY OUTCOME

Extended WaveLinx's commissioning and device-management workflows to support DALI-2 at enterprise scale for one of the most complex hospital lighting deployments in North America. The non-blocking V2 architecture let contractors configure areas **while device import ran in parallel** turning a protocol-imposed wait into invisible background work.

THE SCALE I WAS DESIGNING FOR

10,000+	24	6	128	1,000 ft	250
Devices per hospital floor	Buses per floor	DALI hubs per panel	Fixtures per bus	Max bus length	Devices per hub (per scan)

FIRST DEPLOYMENT Peter Gilgan Mississauga Hospital, the largest DALI-2 installation in North America.

01 · THE PROBLEM

The core problem was time.

Cooper Lighting was retiring two legacy platforms, Fifth Light and iLight. WaveLinx DALI was their DALI-2-certified replacement. There was no product to iterate on, and the DALI-2 protocol's timing rules are physical, they cannot be engineered away.

In V1, adding a hub locked the entire UI behind a blocking loader, *"Processing... Adding DALI Hub and Devices to WAC"* that started at 10 minutes and grew to **27.5 minutes per hub** to absorb 250 devices. Multiply across 10 hubs per controller, and contractors sat idle for hours.

A blocking import, by design

Discovery triggered a full scan of all four buses. No navigation, no area creation, no parallel work, and no progress visibility, until it finished.

Built for 30–50 devices

Every screen, device trees, area lists, the Operate floor plan, assumed a handful of wireless devices. DALI-2 brought thousands.

Make-or-break, not polish

Channel partners and facilities teams were judging whether WaveLinx DALI could credibly replace their tools. A blocking experience at hospital scale was not acceptable.

02 · MY ROLE & SCOPE

UX owner, end to end.

I led UX strategy and design from discovery through handoff, the primary design voice for the entire WaveLinx DALI commissioning system, working directly with Engineering, Architecture, Product, Firmware, and Development, and presenting milestones to groups of 25+ stakeholders for alignment.

What I owned

- Hub discovery, Lobby, Manage Hub, Area UX
- V1 and V2 flow architecture
- Custom iconography (Illustrator → SVG/Fontello)
- Press-ready DALI-2 hub hardware label
- CLS Design System contributions in Figma
- Cross-functional reviews & launch follow-up

DIRECT TEAM

Eng Managers ×2 · Product Owners ×2 · PMs ×2 · Architecture ×1 · Developers ×5

THE KEY CONSTRAINT

The DALI bus scan is slow by protocol design. Addressing 128 devices per bus and fetching their parameters takes real time.

This was never a solvable engineering problem, the protocol ceiling is physical. The UX had to absorb the constraint and design **around** it, not through it.

03 · RESEARCH

Five rooms, one mental model.

Research ran in parallel with design while hardware and firmware were still being built. I used the engineering team as a primary source, supplemented by field context from channel partners and a teardown of three competing DALI commissioning tools. Structured discovery ran across five stakeholder groups before any design started.

01
Engineering & Firmware
Bus bandwidth limits, hub-discovery latency, and the max safe polling frequency during live commissioning.

02
Developers
Loading 3,000+ devices into the existing device tree would time out and degrade the whole app.

03
Architecture & PM
Hospital deployment context, commissioning sequence, and wired-only regulatory requirements.

04
Field Commissioning
Technicians cross-reference floor plans with MAC lists, work bus-by-bus, and need instant device ID under time pressure.

05
Hospital Facilities
Adding devices during construction expansions, rebooting hubs without disrupting live floors, fast fault ID.

Key research insight

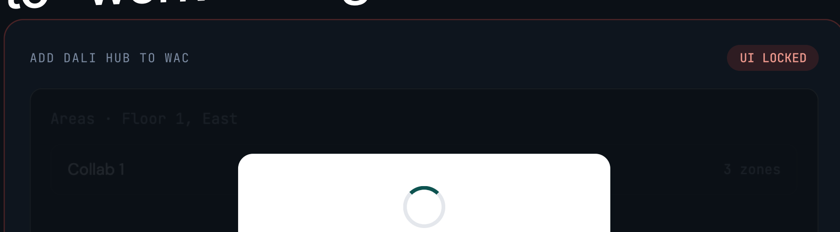
The existing architecture assumed 30–50 devices per controller. **Every** piece of UI was built at that scale. DALI-2 didn't add a feature, it forced a re-examination of how the entire application handled volume.

> *"I've been doing DALI for 12 years. I know what order things need to happen. But your app doesn't, so I have to fight it."*
>
> Master electrician · field research participant

04 · THE PIVOT

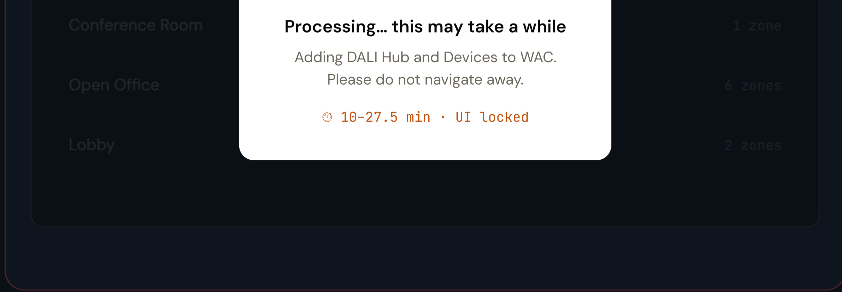
From "wait for it" to "work through it."

V1 · Abandoned | V2 · Shipped



V1, abandoned

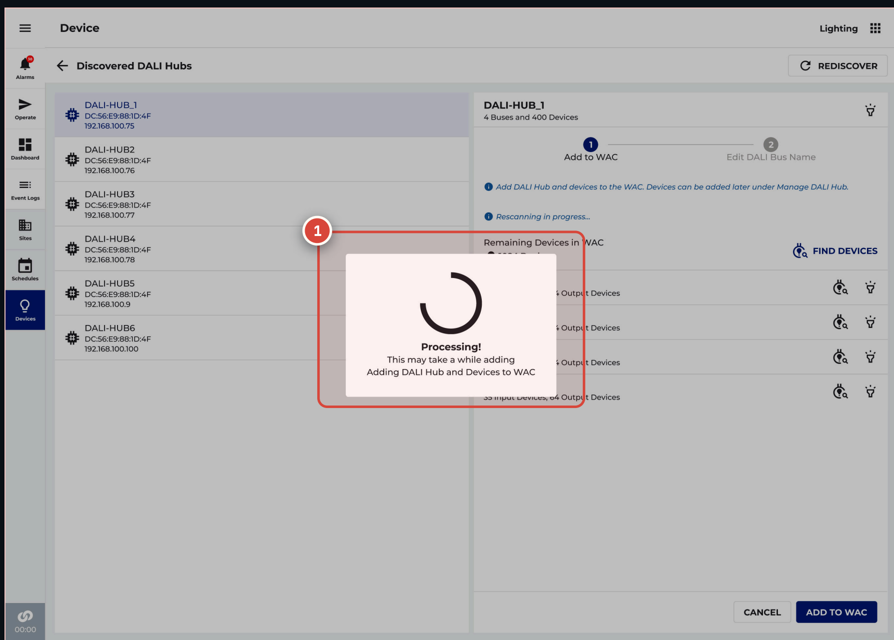
Discovery scanned hubs AND devices at once. Adding a hub locked the UI behind a blocking loader: no navigation, no area creation, nothing, for 10 to 27.5 minutes per hub.



- UI fully blocked during import
- No progress visibility, no fallback
- Contractors idle for hours at scale

THE REFRAME, IN ONE LINE

"We stopped designing for when the scan finishes, and started designing for what users can do while it's still running."

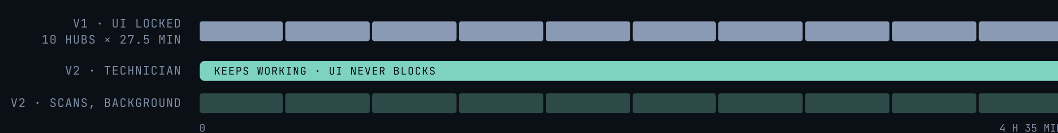


THE REAL V1 ARTIFACT

The actual V1 screen, *"Processing… Adding DALI Hub and Devices to WAC."* It sat on top of a frozen interface for 10–27.5 minutes per hub. V2 deleted this blocking state entirely.

1 Blocking modal, no navigation, no progress, no parallel work.

THE SAME WALL CLOCK, TWO EXPERIENCES



ONE CONTROLLER · WORST CASE DERIVED FROM THE STATED FIGURES (10 HUBS × 27.5 MIN)
GRAY = LOCKED · TEAL = WORKING · PALE = SCAN RUNNING BEHIND THE UI

05 · THE COMMISSIONING FLOW

Step through it. Hub-first import takes a raw hub from the network to a named, addressable system, with a persistent device counter the whole way.

1 **Discover** | 2 Add to WAC | 3 Name | 4 Find devices | 5 Operate

DISCOVERED DALI HUBS · SCAN

- DALI-HUB_1 192.168.100.75
- DALI-HUB2 192.168.100.76
- DALI-HUB3 192.168.100.77
- DALI-HUB4 192.168.100.78
- DALI-HUB5 192.168.100.9

STEP 1 · DISCOVER

Scan the network

REMAINING IN WAC
1,024

From the Device screen the technician runs a network scan. Every DALI hub surfaces automatically, by name, MAC, and IP, ready to import. No IP hunting, no DIP switches.

Network scan complete 10%

BUS 1 — | BUS 2 — | BUS 3 — | BUS 4 —

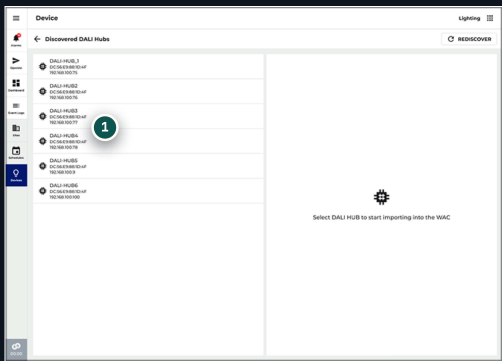
DALI-HUB6
192.168.100.100

awaiting import | awaiting import | awaiting import | awaiting import

Back

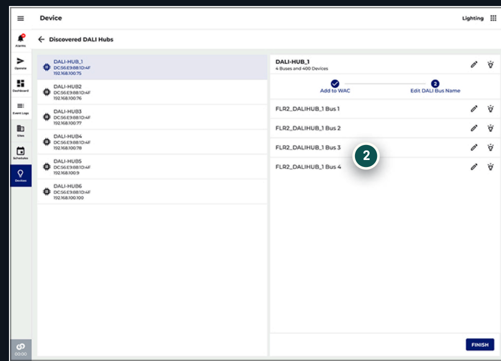
Next step →

SHIPPED TO THE FIELD, THE REAL SCREENS



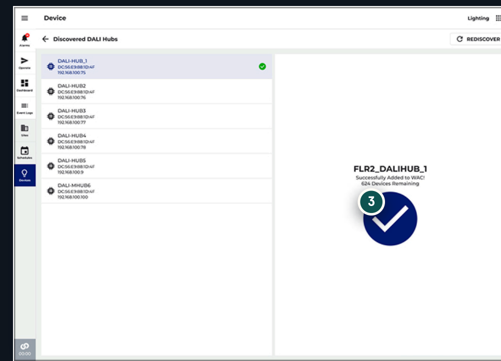
STEP 1 · DISCOVER

Every hub on the network, name, MAC, IP, found automatically.



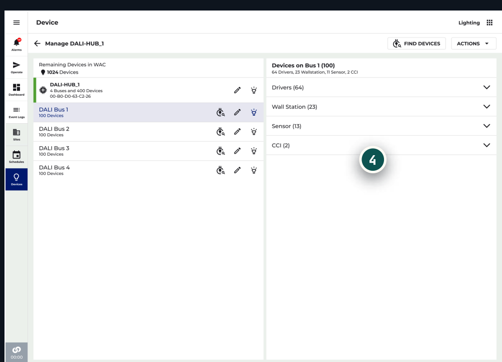
STEP 2 · NAME

Rename hub and buses inline, tied to floor and location.



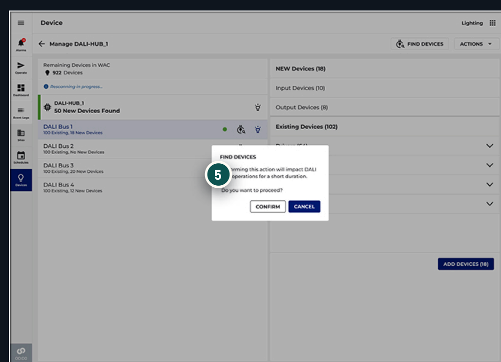
STEP 3 · ADDED TO WAC

Confirmation with the remaining-device count, on to the next hub.



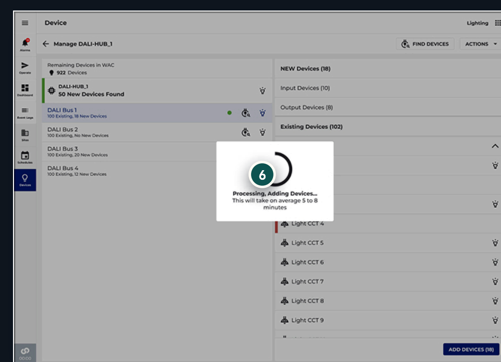
MANAGE HUB

Live per-bus counts: Drivers 64, Wall Stations 23, Sensors 13, CCIs 2.



FIND DEVICES

A confirm dialog guards a live hub before any rescan runs.



ADD DEVICES

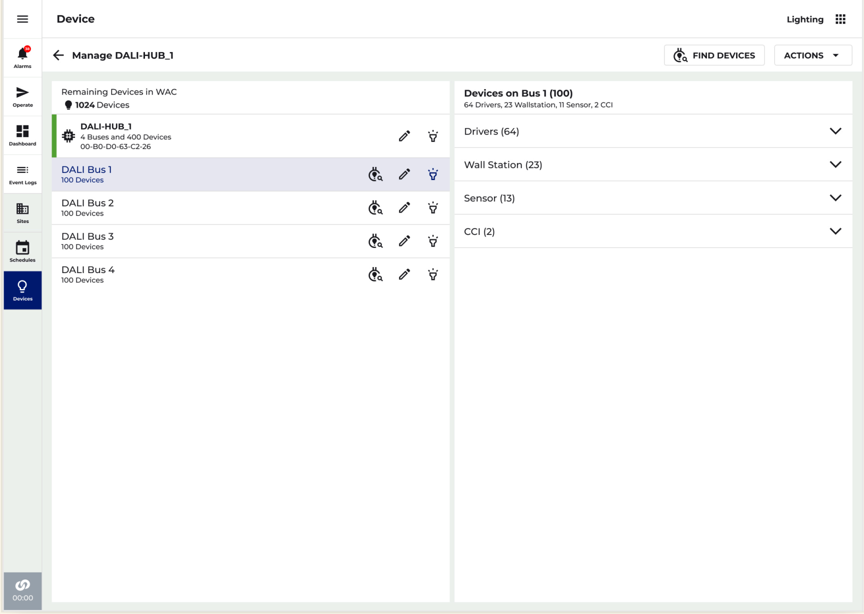
A processing modal sets a 5–8 min expectation, then a toast confirms.

06 · DALI HUB MANAGEMENT

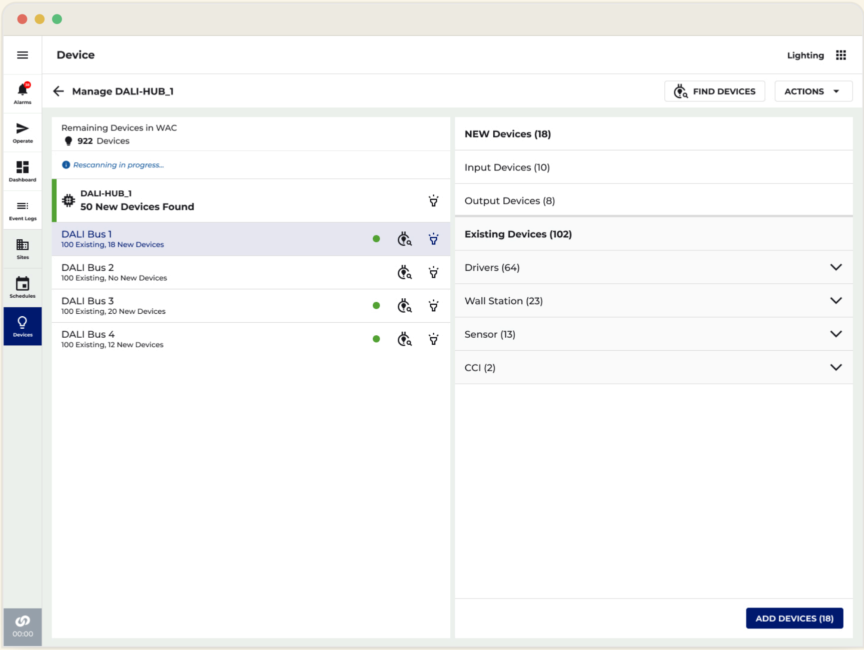
Commissioning is only the beginning. Hospitals expand, floors get reconfigured, and devices get added to existing buses over time. The Manage Hub screen supports the entire lifecycle of a DALI-2 install, letting technicians monitor hub health, rescan for new devices, and take corrective action without disrupting a live facility.

MANAGE HUB

The left panel lists every bus under the hub with live device counts. Selecting a bus breaks inventory down by category, Drivers (64), Wall Stations (23), Sensors (13), CCIs (2). A persistent **Remaining Devices in WAC** counter keeps installation-wide context in view.



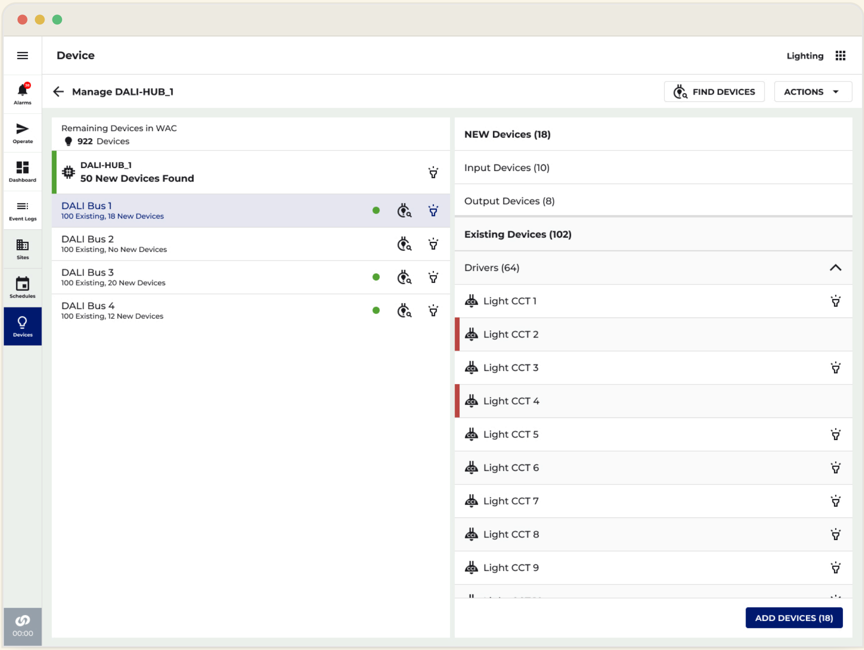
ONE BUS AGAINST THE 128-DEVICE PROTOCOL CEILING · 102 USED, TOTAL DERIVED FROM THE STATED COUNTS



FIND DEVICES

When devices are added to the physical install post-commissioning, the technician triggers Find Devices from the top bar. Because the scan temporarily impacts hub operations, a confirmation dialog protects against accidental triggering on a live system.

Buses with newly discovered devices are flagged with a **green indicator** and a new-vs-existing count, so technicians know which buses need attention without scanning each one manually.



ADD DEVICES

The detail panel separates **New Devices** from **Existing Devices**, organized by type. A single ADD DEVICES (n) action imports all of them. A processing modal sets expectations with a 5–8 minute estimate, and a toast confirms the result: "18 NEW Devices added to Bus 1."

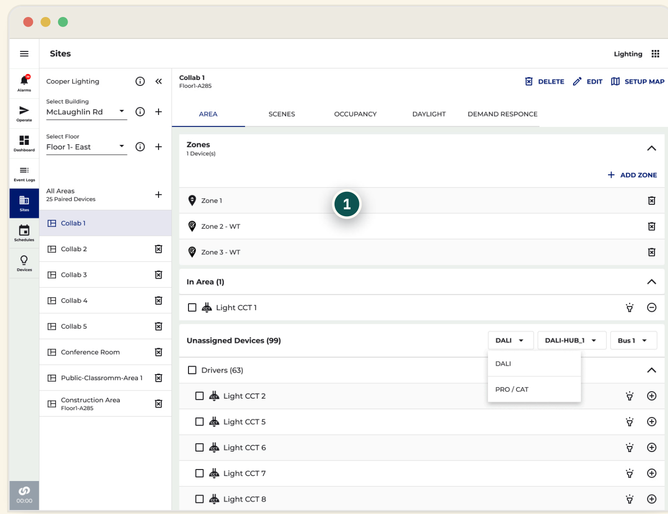
UX DESIGN DECISION

Separating new from existing devices before committing any change was a deliberate safeguard at hospital scale, blindly importing an unexpected count could cause real downstream commissioning problems. The green bus indicators give at-a-glance triage so technicians can prioritise which buses to address first, and the processing modal's time estimate was added in direct response to commissioning teams who needed to plan around the 5–8 minute rescan window on a live hub.

07 · AREA COMMISSIONING

A hospital floor with six-plus hubs could dump thousands of devices into the Unassigned list, flooding the screen and crippling load times. So I inverted the default: instead of "show everything," the technician picks a scope first.

A **"Select to Start"** empty state with System (DALI or PRO/CAT) → Hub → Bus dropdowns narrows the list to a single bus's 128 devices before anything renders. It reframes the work as bus-by-bus, exactly how DALI-2 is physically wired.



1 The System → Hub → Bus selectors scope the Unassigned list to a single bus's 128 devices before anything renders.

Scoped to 128 devices per bus, not the full 10,000.

08 · FLOOR PLAN VIEW

The fixed device tree became a Building → Floor → Area → Zone → Device dropdown bar, and the map was clustered. **Zoom in** to break clusters into devices; toggle type or health.

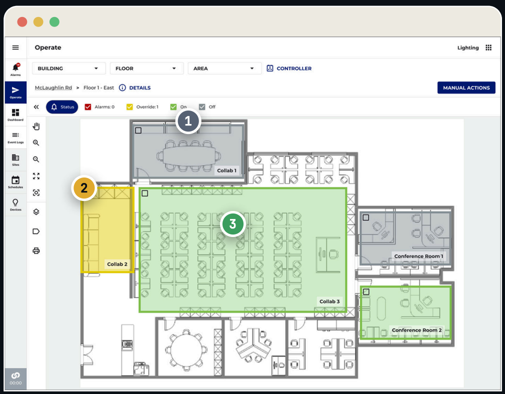
McLaughlin Rd ▾ | Floor 1, East ▾ | Area ▾ | Zone ▾

Clustered | Device level

McLaughlin Rd › Floor 1, East › Conference 1

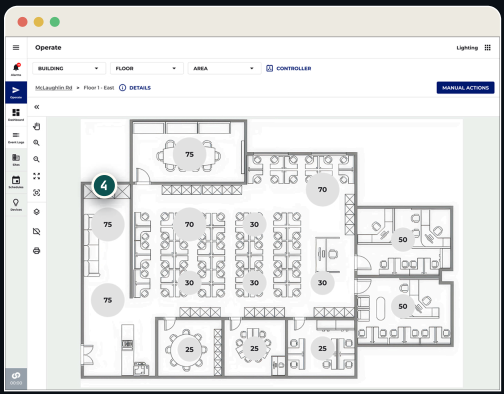
12 clusters · Site zoom

THE SAME MAP, EVERY SURFACE



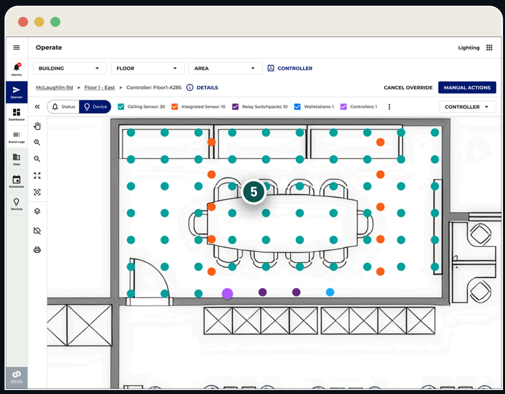
1 Collab 1 · off 2 Collab 2 · override 3 Collab 3 · on

Web · floor-wide health status, rooms color-coded by state.



CLUSTERED

Device counts per area, readable at floor zoom.

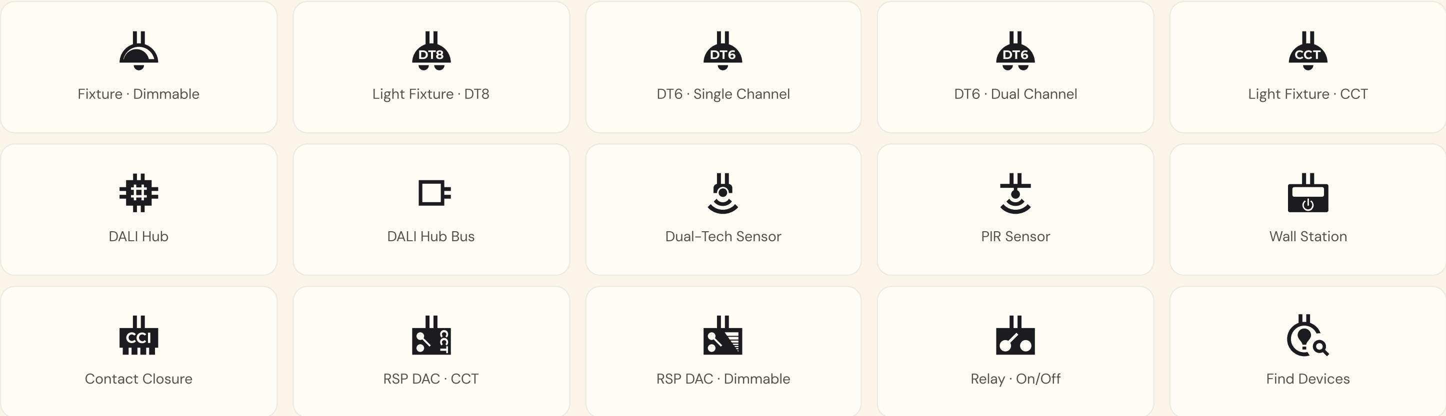


DEVICE LEVEL

Zoom in: individual devices, color-coded by type.

09 · CRAFT, A VISUAL LANGUAGE

I custom-designed the device icon set in Adobe Illustrator, embedding the DALI 2-wire bus, the defining physical trait of every DALI device, as a shared motif. Users recognize "this is a DALI device" as a category, then read the type at a glance. Exported as SVG, validated for rendering via Fontello.

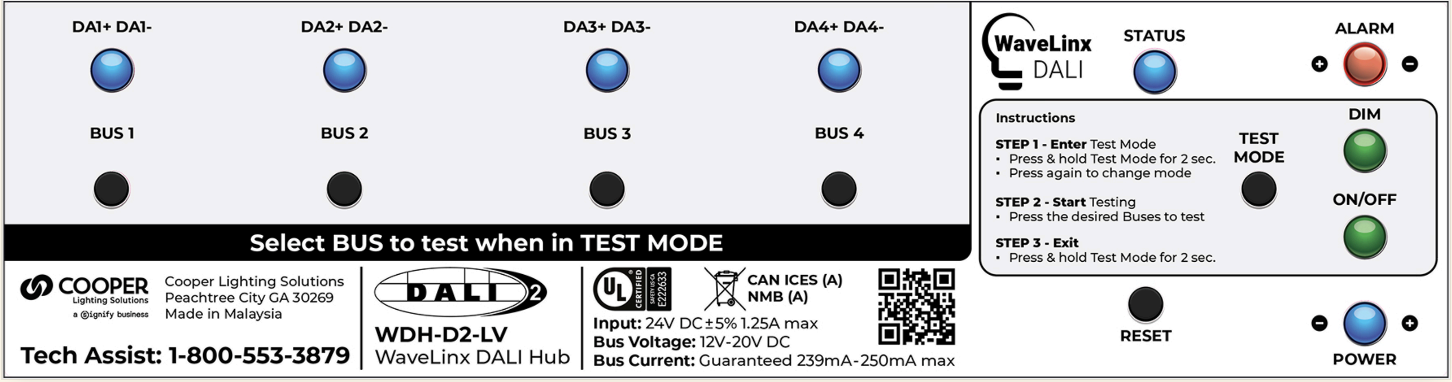


Fixture · Dimmable	Light Fixture · DT8	DT6 · Single Channel	DT6 · Dual Channel	Light Fixture · CCT
DALI Hub	DALI Hub Bus	Dual-Tech Sensor	PIR Sensor	Wall Station
Contact Closure	RSP DAC · CCT	RSP DAC · Dimmable	Relay · On/Off	Find Devices

Press-ready hardware label

I also designed the WDH-D2-LV hub label, die-cut, spec-complete, with test-mode instructions, created in Illustrator and shipped on the hardware.

ILLUSTRATOR · PRESS-READY PDF



10 · ITERATING IN AMBIGUITY

Hardware prototypes arrived on a rolling basis and third-party device availability was uncertain throughout. Weekly cross-functional reviews kept design in lockstep with engineering. Each challenge below maps to a concrete design response.

PHASE	CHALLENGE	DESIGN RESPONSE
V1 → V2 pivot	A UX review exposed the 10-minute blocking import as structurally unacceptable at scale.	Reframed from "scan then configure" to "add hub and configure while the scan runs in background." Final timeout 27.5 min, non-blocking.
Manage Hub 3-step	The initial Manage Hub flow was an undifferentiated list; field and engineering flagged it as unclear.	Defined a 3-step flow: rescan & find, show types & identify, add devices, with confirmation and cancel at each step.
Device volume / floor plan	The existing floor-plan hierarchy would have crippled load times at DALI-2 scale.	System-wide UX stress-test; introduced dropdown navigation, clustering, and filter-before-load to handle 3,000+ devices.
DT8 dual-channel zones	A single DT8 driver presents as one device but needs two zone types, Dimmable and White Tunable.	Defined an auto-select rule and multi-zone pattern mirroring the 2-channel PRO Node model.
Naming & icons	DT6, DT8, DAC, Field Relay and CCI in the same lists needed consistent naming and taxonomy before QA could test.	Custom two-wire icon system and a full naming convention; DALI-1 and DALI-2 DT6 confirmed to render identically.

11 · OUTCOMES

0+

0/12

0s

Devices at full hospital scale

LEGACY 30–50 → 10,000+ · ONE SCALE

QA regression scenarios passed

12 OF 12 PASSED

0→100% light transition, all device types (CTQ)

0→100% WITHIN 2.5 S · EVERY TYPE

`0 → 1 net-new product category` `Largest DALI-2 install in North America` `200+ device types validated across 4 buses`

- Hub-first, non-blocking commissioning across the mobile app
- DT8 dual-channel (CCT / white-tunable) driver support with auto-addressing
- Filter-before-load area commissioning, scoped to a single 128-device bus
- Custom DALI-2 icon system with a shared two-wire motif
- Identify at hub, bus, and device level for physical-to-logical verification
- Mixed-protocol zones: DALI, CAT wired, and WaveLinx wireless together

THE HARDWARE IT SPEAKS TO

SENSE

Dual-tech & PIR sensors

Occupancy & daylight, per zone

POWER

Driver & relay panels

Up to 128 fixtures per bus

CONTROL

Scene wallstation

Recall scenes, raise & lower

WHY THIS WORK NEEDED MY BACKGROUND

The V1→V2 pivot meant recognizing that a 10-minute blocking import wasn't a loading-state problem, it was an architectural mismatch between the commissioning mental model and the protocol's real timing. The engineering answer was a longer timeout. **The UX answer was to make the entire wait invisible by designing around it.**

Hardware bandwidth limits, device naming, the dual-channel abstraction, the floor-plan hierarchy, none of these were decisions made in isolation. Each required navigating engineering constraints, product commitments, QA dependencies, and hardware availability at once. My experience across connected lighting, enterprise SaaS, and hardware-adjacent environments gave me the credibility and domain fluency to lead these calls rather than react to them.

Design-system leverage

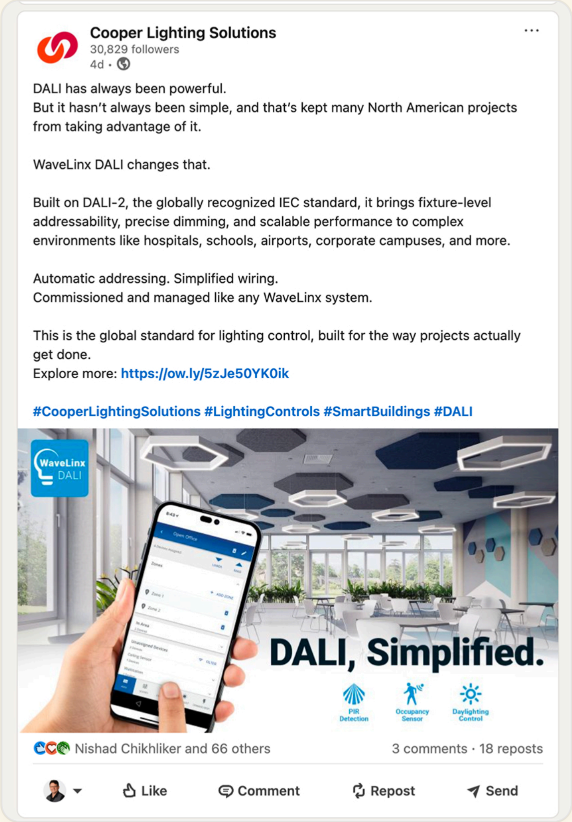
The patterns from DALI, non-blocking scan, the two-banner progress model, filter-before-load, dropdown navigation, device-vs-endpoint display logic, are now the reference architecture for future IoT subsystems in WaveLinx, documented in the CLS Design System for reuse.

12 · THE ECHO

After release, Cooper Lighting's own LinkedIn campaign led with the exact promise this work was built around, **"DALI, Simplified."** Automatic addressing, simplified wiring, commissioned like any WaveLinx system, and the campaign's hero image is the shipped commissioning app itself.

When marketing repeats the simplification story that UX orchestrated across product, firmware, engineering, and field teams, the design has stopped being an internal argument and become the product's public voice.

> *"DALI has always been powerful. But it hasn't always been simple — WaveLinx DALI changes that."*
>
> COOPER LIGHTING SOLUTIONS · LINKEDIN CAMPAIGN



The campaign creative is the shipped commissioning UI.

ANOTHER CASE STUDY

CORE Insights →

rbabiarz@gmail.com

← Back to portfolio

COLOPHON · BUILD RECORD

VIEW SOURCE ON GITHUB ↗

SITE EST	DEPLOYMENTS	LAST DEPLOY	VERSION	UPDATED
Jun 22, 2026	108	Jul 18, 2026	v1.11.113	Jul 19, 2026

BUILT WITH — Anthropic Claude Code · Playwright · Figma + MUI kit · Cursor · Python 3 · React 18 (CDN, no build step) · Git → GitHub Pages

ARCHITECTURE — 29 self-contained pages, ~76,000 hand-authored lines, zero npm dependencies, no bundler — every page is a single file that opens anywhere and deploys as pure static hosting. The interactive layer is bespoke vanilla JS: a photometric engine scored against real parking uniformity targets, an AI AutoLayout demo with ranked candidates, playable decision drills, and client-side search over a Python-generated index. WCAG 2.2 AA target with a standing audit; three-tier design tokens (primitive → semantic → component).

AI-PAIRED — Human-led, AI-accelerated: designed and engineered in pair with Anthropic Claude Code, with Playwright browser automation verifying every change — the same receipts-over-vibes workflow the case studies argue for. 11 case studies, several of them about designing AI systems people can trust.

STATS DERIVED FROM GIT HISTORY & THE GITHUB PAGES DEPLOYMENT API, REGENERATED PER RELEASE · VERSION = GENERATION · CASE STUDIES · COMMITS